Tensor computations

Monotonicity of the average number of critical rank-one approximations

Difficulty: challenging **Importance:** interesting to the community

Statement

For integers $p \geq 3$ and $2 \leq n_1 \leq \cdots \leq n_p$, let $G \in \bigotimes_{i=1}^{p} \mathbb{R}^{n_i}$ have independent standard normal entries. Let X be the smooth manifold of nonzero real rank-one tensors in this space, and let $N(G)$ be the number of critical points on X of

$$Z \longmapsto \|G - Z\|_F^2.$$

Each critical tensor is counted once, regardless of how its vector factors are scaled. Define $a(n_1, \ldots, n_p) = \mathbb{E}[N(G)]$; the critical-point count is finite almost surely.

If

$$n_p - 1 > \sum_{i=1}^{p-1} (n_i - 1),$$

is it always true that

$$a(n_1, \ldots, n_{p-1}, n_p) \leq a(n_1, \ldots, n_{p-1}, n_p - 1)?$$

The expectation on the right uses a standard Gaussian tensor in the smaller space.

Relevance

Rank-one approximation algorithms navigate a landscape of stationary points. This conjecture predicts how the average number of such points changes when one mode becomes much larger than the others.

References

1. J. Draisma and E. Horobeţ, *The average number of critical rank-one approximations to a tensor*, Linear Multilinear Algebra 64 (2016), 2498–2518. DOI; primary preprint, Conjecture 1.3, p.3; §1 defines the Gaussian average and Theorem 1.1 gives an integral formula.
2. S. Friedland and G. Ottaviani, *The Number of Singular Vector Tuples and Uniqueness of Best Rank-One Approximation of Tensors*, Found. Comput. Math. 14 (2014), 1209–1242. DOI; primary preprint, Theorem 1, for the corresponding complex critical-point count.

Status check — 2026-09-08

The latest originating arXiv revision (November 2, 2015) retains Conjecture 1.3. Searches *"critical rank-one" "monotonicity"*, *"Draisma" "Horobet" "Conjecture 1.3"*, and *"Draisma" "Horobet" "conjecture" proof stabilization* located no proof or counterexample. Constancy results for the complex count do not settle this real Gaussian expectation. No recent primary reaffirmation was located.